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Assignment-2

Q1 - What is Redux in React?

Ans:- Redux is a popular state management library used with React applications. It stores the application data in a single central store. Any component can access or update the data without passing props manually through multiple components.

Redux helps in solving the problem of prop drilling and makes data flow easier to manage.

Q2 - Why Redux is Used?

Ans:- → Centralized state management.

→ Easy data sharing between components.

→ Predictable state updates.

→ Better debugging.

→ Avoids prop drilling.

Q3- What are Main Components of Redux?

Ans:- 1. Store:

The store keeps the complete state of the application.

2. Actions:

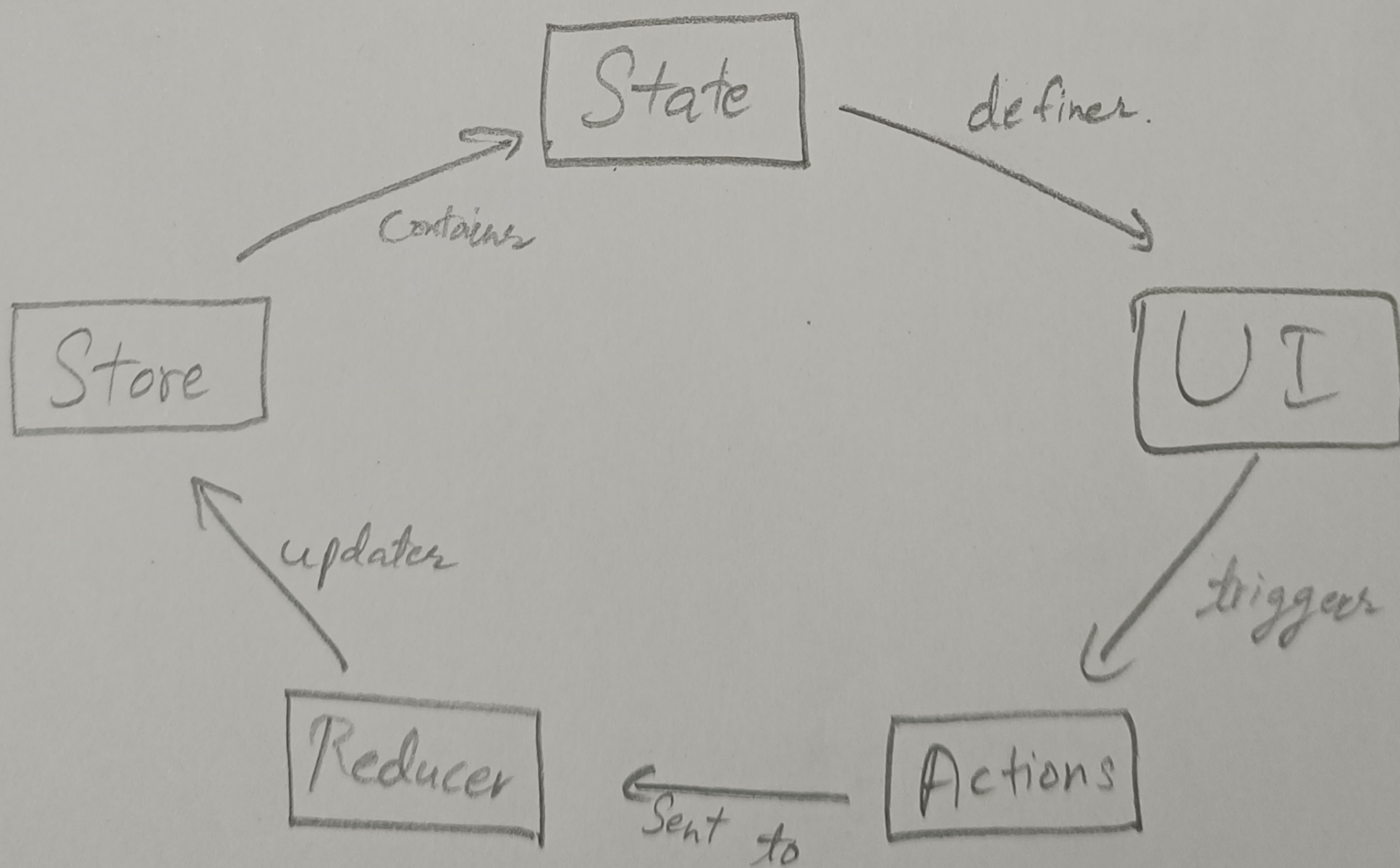
Actions are objects that describe what operation to perform such as increment or decrement.

3. Reducer:

Reducer is a function that updates the state according to the action received.

4. UI Components:

React components display data and trigger actions.



Q4- Program Description

Ans:-

- Grandparent component displays count value.
- Grandchild component contains Increment & Decrement buttons.
- Clicking Increment increases count.
- Clicking Decrement decreases count.
- State is managed using Redux store.

* Advantages of Redux:

- (i) Grandchild can update data directly.
- (ii) Grandparent can display updated value directly & instantly.
- (iii) No need to pass props through Parent & Child components.